

The Answer–Mark Gap: MYP Vault

An AI Study Assistant Built to Close the Gap Between Revision and Marks · Anvitha Abbagani · 2026

IB students spend hours revising but consistently lose marks on the same question types; not because they don't know the content, but because they can't translate knowledge into mark-scheme-aligned answers. MYP Vault is a RAG-based AI assistant that grounds every response in real mark schemes, giving students structured, scoreable answers instead of generic explanations.

Full supporting detail – competitive analysis, complete failure-mode breakdown, test dataset, system prompt, wireframes – is in the Appendix.

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1. Problem

Who: MYP students (Grades 6–10) preparing for IB exams, studying Biology, Math, and Psychology through past papers, ChatGPT, and scattered websites – switching between 2-5 tools per session with no reliable feedback on whether their answers would actually score.

Where it breaks: A student attempts a past paper question, compares it to the mark scheme (which assumes examiner-level interpretation), pastes it into ChatGPT for a well-explained but unstructured answer, tries to reconcile the two, and retries without ever knowing exactly what cost them the mark.

"I understand the topic, but I don't know how to write the answer to get full marks." – Grade 10 student, switches between 4 tools per session

Research: 12 discovery interviews (8 MYP students, 2 IBDP students, 2 parents). Every interviewee pointed to the same root cause: not a knowledge gap, a translation gap. Students already use ChatGPT – just for the wrong job (grading, not generating mark-scoring answers). MYP Vault redirects existing behavior rather than creating new behavior from scratch.

Why RAG, not fine-tuning or prompting:

Approach	Why / Why Not
Prompt Engineering	No access to IB-specific sources → generic answers
Fine-Tuning	Static knowledge → outdated the moment mark schemes update each exam cycle
RAG (chosen)	Retrieves real mark schemes → grounded, current, trustworthy

Competitive positioning (full analysis: Appendix A): No competitor– not Revision Village, not general AI tools –solves the translation gap. That's the wedge, and it's why the North Star metric is Answer Score Improvement Rate, not engagement: it's the one thing a content-only competitor can't fake.

2. Product Decisions

P0 user stories (the product doesn't exist without these):

1. Structured, one-bullet-per-mark answers for exam questions
2. Answer evaluation showing exactly which mark scheme points were hit vs. missed
3. Unified retrieval – question, mark scheme, and concept in one query

Mark-scoring generation and unified retrieval scored highest and got built first; mistake tracking, follow-up questions, and confidence indicators were explicitly deprioritized – retention features are worthless until the core answer-generation loop is trustworthy. Full P1/P2 stories and RICE prioritization: Appendix B.

Why I didn't build a full study platform: Flashcards, quizzes, and tutoring were cut from v1 by design. Students don't lack content, they lack a way to convert it into mark-scoring answers; building flashcards

would have addressed a secondary problem while delaying the primary one. Phase 2 features unlock only when >75% of active users show measurable Answer Score Improvement – until that number moves, added scope just adds surface area.

Architecture – three decisions that mattered most (full pipeline: Appendix C):

1. **1 mark = 1 chunk**: First attempt chunked mark schemes as full paragraphs. The model merged multiple marking points into one response, so a student could get a correct-sounding answer that scored 2/4 with no way to see what was missing. Moving to one-mark-per-chunk fixed this at the cost of reading fluency. Decision: scoring precision beats fluency, because IB grades per-point.
2. **Retrieve 12, filter to 6**: K=5 missed key mark scheme points; K=10 introduced noise. K=12 filtered to 6 was the best balance, reflecting the core asymmetry: for this product, missing a mark scheme point is a worse failure than surfacing one redundant chunk.
3. **GPT-5 Mini over Claude/Gemini**: All three tested comparably on accuracy. GPT-5 Mini won v1 on cost-to-performance (\$0.25/\$2.00 per 1M tokens) with sufficient image support and instruction-following for structured grading. Claude Sonnet 4.6, while strong on explanations, is cost-prohibitive at prototype stage at \$3.00/\$15.00 per 1M tokens (input/output). Prototyped but not shipped: Gemini 2.5 Pro (\$1.25/\$10.00 per 1M tokens) was tested for Explanation Mode (Appendix H) and produced strong analogy-based explanations; held for v2 pending usage data that justifies the added cost, while keeping GPT-5 Mini for Exam and Evaluation modes – Gemini's pricing is roughly 2.4x cheaper on input and 1.5x cheaper on output than Claude's, giving a blended model a meaningfully lower cost floor than running Claude Sonnet directly.

3. Failure Modes & Metrics

RAG systems fail in predictable places. Three failure modes mattered most for this product (full breakdown of all 9: Appendix D):

Failure	What Breaks	Fix	Decision
OCR / data quality	Noisy scans → wrong marking points → wrong scoring	Preprocessing pipeline + manual validation on high-mark, high-frequency subjects	If the source is wrong, the system confidently gives wrong answers. This breaks trust fastest, so it gets prioritized over full-corpus coverage
Chunking	Split incorrectly → system loses the question→mark→score mapping	1 mark = 1 chunk	Scoring precision over readability
Retrieval	Only some required mark scheme points retrieved → incomplete answers	Hybrid retrieval (semantic + metadata) + K=12→6	Recall over precision – better to over-surface than under-surface a mark

The remaining six (embedding quality, re-ranking, prompt/augmentation, model quality, data drift, user behavior) follow the same pattern: detect via a quantified threshold, fix, and re-affirm the same core trade-off of **accuracy over polish**.

North Star: Answer Score Improvement Rate – the % of users who score higher on the same question type after 2+ sessions. This is deliberately not an engagement metric; it measures whether revision is actually converting into marks, which is the entire premise of the product.

Guardrails: Hallucination rate (target <10%), 0% copyright violations, 100% correct out-of-scope refusal. Evaluation method: human audit against real mark schemes + LLM-as-judge, tested against a 10-question set spanning Exam/Evaluation/Explanation modes (full set: Appendix E).

4. Economics, Rollout & What's Next

Cost: ~3,813 tokens/query on GPT-5 Mini $\approx$ \$0.0022/query $\rightarrow$ ~\$1.32/user/month at normal usage, ~\$2.64 during exam-season spikes (2x usage).

Pricing: Free tier (25 queries/day) $\rightarrow$ Pro tier (\$8/month, unlimited + mistake tracking). Gross margin 83.5% normal, 67% during exam season. Break-even at ~15 paying users against \$100/month fixed infrastructure cost. The free-tier cap is intentional: students hit it during active revision, exactly when the product's value is most felt.

Launch: Phase 1 (small dataset, gate on retrieval precision >70% and hallucination <10%) $\rightarrow$ Phase 2 (real student testing, gate on positive feedback >70%) $\rightarrow$ Launch (scale, add retention features). GTM is community-led: r/IBO and r/IBMYP demos, a school ambassador program (1 early-access seat per 5 referrals), and teacher outreach (one teacher endorsement $\approx$ 30 students). Target: 100 active users in 30 days before any paid acquisition.

What's next:

- Fine-tuning: Once evaluation data is standardized, use PEFT/LoRA to reliably classify mistake types (concept gap vs. structure vs. missing keywords), with a train-large/distill-small strategy to keep production cost low.
 - Agentic RAG: Multi-step study agents that generate a full study plan, track weak areas, and proactively suggest revision tasks – gated behind a working core loop.
 - Parent-facing mode: Discovery interviews surfaced a secondary segment – parents funding tuition who understand the content but can't translate it into IB-aligned explanations for their kids. High willingness to pay, natural expansion once the student product is validated.
 - Teacher accounts: a separate login for teachers to upload their own resources (e.g., presentations) for students – scoped out of v1 pending review of shared-content permissions and moderation.
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Appendix

A: Competitive Analysis

Product	Approach	Strength	Gap vs. MYP Vault
Revision Village	Video lessons + question bank, subscription	Strong content library, established IB brand	No answer generation or evaluation. Students still self-mark with no feedback on why points were lost
ChatGPT / Gemini	General LLM, no retrieval grounding	Already in every student's workflow, free, flexible	Not grounded in mark schemes. well-explained but ungraded, unstructured
Study websites	Static reference content	Free, broad coverage	Unstructured, no personalization, no feedback loop
MYP Vault	RAG grounded in real mark schemes	Structured, scoreable, per-mark feedback, single tool	N/A

MYP Vault's own trade-off: smaller content library at launch, since v1 depends on user-uploaded papers rather than a pre-loaded corpus.

Positioning: Revision Village wins on content volume. MYP Vault wins on answer precision. These aren't competing for the same use case – a student could use Revision Village to review content, then MYP Vault to practice scoring higher on it. That's part of why the GTM plan targets the same communities Revision Village grew from instead of pitching MYP Vault as a replacement.

Why this matters: No competitor solves the translation gap of knowing content vs. writing an answer that scores marks. Copying it means rebuilding the same mark-scheme-grounded RAG pipeline, not just shipping a new UI. This is why Answer Score Improvement Rate, not engagement, is the North Star: it directly measures the one thing a content-only competitor can't fake.

B: Full User Stories & Prioritization

P0 (Core Value)

1. As an IB student attempting a 4-mark Biology question, I want a structured answer with one bullet per mark, so I don't have to interpret the mark scheme myself.
2. As an IB student who just submitted an answer, I want to see exactly which mark scheme points I hit and missed, so I can correct the specific gap.
3. As an IB student mid-revision, I want mark scheme points, concept explanations, and past paper questions in a single query, so I don't switch tools.

P1 (Retention Drivers)

4. As an IB student burned by inaccurate AI answers before, I want every answer to cite its source, so I can verify rather than trust blindly.

5. As an IB student two weeks from exams, I want to see my repeated mistake patterns, so I can prioritize the corrections that recover the most marks.
6. As an IB student in Explanation Mode, I want suggested follow-up questions, so I can test my understanding under exam conditions.

P2 (Optimization)

7. As an IB student, I want to know if a topic is still in the syllabus, so I don't waste time.
8. As an IB student, I want a confidence indicator, so I know whether to revise further.

RICE Prioritization

Feature	Reach	Impact	Confidence	Effort	Score
Mark-scoring answer generation	3	3	3	2	13.5
Unified resource retrieval	3	3	3	2	13.5
Instant answer feedback	3	2	2	1	12
Source grounding + citations	3	2	3	2	9
Topic-based question retrieval	3	2	2	2	6
Mistake tracking	3	2	2	2	6
Re-ranking (exam relevance)	2	2	2	2	4
Follow-up questions	2	2	2	2	4
Syllabus relevance detection	2	1	1	2	1
Confidence indicator	2	1	1	3	0.67

Negative use cases:

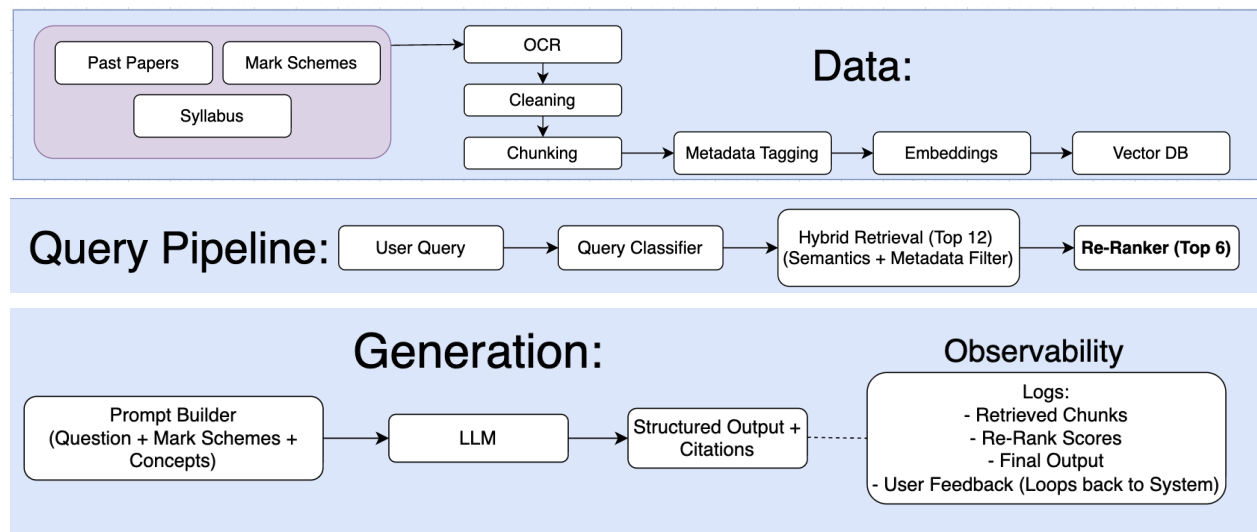
- Out-of-scope queries ("Explain black holes," "Write my Economics IA") → politely refuse, redirect to IB scope.
- Full past paper / mark scheme requests → refuse full derivation, provide extracted relevant parts and summaries only.

C: Technical Architecture (Full Pipeline)

Data sources & priority: Past papers (High, questions) · Mark schemes (Critical, scoring) · Syllabus (Medium, relevance)

Ingestion: OCR → cleaning (remove headers/footers/watermarks – noisy text produces worse chunking and worse answers) → chunking → metadata tagging → embeddings → vector DB.

System Diagram (RAG Pipeline):



Chunking strategy

Chunk Type	Strategy	Why
Question	Full unit	Preserve intent
Mark scheme	1 point = 1 chunk	Enables per-mark scoring
Syllabus	Topic-level	Enables filtering

Retrieval pipeline: Query classified into Exam / Evaluation / Explanation Mode → hybrid retrieval (semantic search + metadata filtering on subject/topic/marks/command term) retrieves the top 12 candidate chunks from past papers and mark schemes → re-ranked by mark scheme priority, command term match, mark allocation match, topic relevance → top 6 sent to the model, typically 1-2 question chunks and 3-6 mark scheme chunks. For Explanation Mode, the model draws on syllabus context plus its own subject knowledge to generate concept explanations – concepts aren't a separately ingested corpus.

Top-K testing: K=5 (low recall, missed mark scheme points) → K=10 (noise from tangential mark scheme chunks outside the question's topic) → K=12 filtered to 6 (best balance).

Context window budget: ~1,250 words per call (900 words retrieved context + 300 system prompt + 50 query) ≈ 3,125 input tokens + ~688 output tokens ≈ 3,813 tokens/call.

Why large-context (full past paper in context) was rejected: the model struggled to isolate relevant marking points and answers became progressively less aligned to marks as context grew. Constrained retrieval outperformed brute-force context size.

Observability: every query logs retrieved chunks, re-rank scores, final output, and user feedback – feeding back into retrieval tuning. Tracked: missed mark rate, frequency of partial answers, retry rate.

D: Full RAG Failure Mode Analysis (All 9)

Failure	What Breaks	Detection	Fix	Decision
Data Quality / OCR	Noisy OCR → wrong marking points → wrong scoring	Malformed tokens >5%, retrieval precision drop	Preprocessing + manual validation of high-priority mark schemes	Prioritize cleaning high-frequency subjects + high-mark questions
Chunking	Split incorrectly → loses question → mark → score mapping	Output has fewer points than marks allocated	1 mark = 1 chunk	Scoring precision over fluency
Embedding Quality	Misinterprets command terms (define vs. explain)	Correct topic, wrong structure/length	Embed command term + marks, not just topic	Accept latency for exam precision
Search / Retrieval	Only some required points retrieved	Precision <70%, missing expected points	Hybrid retrieval + K=12 → 6	Prioritize recall over precision
Re-ranking	Mark scheme chunks not prioritized over concept notes	Correct chunks retrieved but unused	Increase mark scheme weight in ranking	Scoring over conceptual clarity
Prompt / Augmentation	Too much unstructured context → hallucination	Hallucination rate >10%	Limit to 5-8 high-quality chunks, enforce structure	Concise context over depth
Model Quality	Weak model scores wrong; strong model costs too much	Incorrect scoring / high cost	Mid-tier model default, upgrade selectively	Use stronger models only where needed
Data Drift	Outdated syllabus → irrelevant answers	Increase in outdated answers	Update after each exam cycle	Align updates with exam timelines, not real-time
User Behavior	Vague queries → poor outputs	Missing command terms, high clarification rate	Prompt user for command term + marks	Guide only when necessary

E: Evaluation Test Dataset

Exam Mode

1. Explain peristalsis (4 marks) → definition, muscle action, direction, control
2. Compare plant and animal cells (6 marks) → similarities (membrane, cytoplasm, nucleus, mitochondria, ribosomes) + differences (cell wall, chloroplasts, vacuole, shape, energy storage)

3. Solve $3x + 12 = 9 \rightarrow x = -1$, with steps shown
4. Explain why increasing temperature increases reaction rate (4 marks) $\rightarrow$ kinetic energy, collision frequency, activation energy, successful collisions
5. "Photosynthesis makes food" – score this 4 marker $\rightarrow$ low score (1/4), missing definition/process/inputs-outputs, feedback to add equation
6. Explain advantages of a sole trader (6 marks) $\rightarrow$ control, profit retention, ease of setup, flexibility, customer relationship, fewer legal requirements
7. Define diffusion (2 marks) $\rightarrow$ movement of particles, high to low concentration

Evaluation Mode

8. "Diffusion is movement" (2 marks) $\rightarrow$ 1/2, missing concentration gradient
9. "An enzyme helps reactions" (2 marks) $\rightarrow$ 1/2, missing biological + catalyst

Explanation Mode

10. "I don't understand diffusion" $\rightarrow$ expects analogy + explanation + diagram + exam link

Escalation design: when no relevant chunks are retrieved, the system states uncertainty explicitly ("I'm not fully confident in this answer based on the available mark schemes"), offers the closest related topics, and asks the user to clarify – rather than guessing.

Success criteria: >80% of queries retrieve at least one correct mark scheme chunk · >70% of top-6 retrieved chunks relevant · >75% of answers match mark scheme explanations · 0% copyright violations · 100% correct out-of-scope refusal.

F: Data & Permissions

- Each user's uploaded papers and mark scheme chunks are private and isolated per account via user-ID metadata filtering, alongside personal notes and answer history. Syllabus content is the one resource maintained centrally, since it's curriculum structure, not copyrighted exam material.
- IB content rights: IBO permits past papers for personal study but prohibits commercial redistribution. V1 requires users to upload their own purchased papers – no bulk server-side storage. The system stores mark scheme chunks only, not full questions. Legal review required before expansion.
- Copyright reconstruction risk: a student could attempt to rebuild a full paper by requesting questions sequentially across multiple sessions. Mitigation: track query history for sequential-request patterns across sessions (not just within one) and prevent full-document reconstruction; offer similar questions from other papers instead.
- Account lifecycle: On cancellation, personal data (notes, answer history) is retained at reduced access for 30 days, then downgraded to aggregate stats only. On account deletion, all personal data is permanently purged within 30 days with email confirmation. Resource-level revocation (e.g., a flagged copyright issue) removes content from future retrieval without retroactively altering past answers.

G: System Prompt (Abbreviated)

“You are MYP Vault, an AI assistant specialized in helping IB MYP students prepare for exams using past papers, mark schemes, and IB-aligned resources. Your primary goal is to maximize student marks by generating responses that strictly reflect mark scheme expectations.

Query Classification: Exam query (command term + marks) → Exam Mode · Evaluation query (student submits own answer) → Evaluation Mode · Doubt query (no marks, expresses confusion) → Explanation Mode.

Exam Mode: bullet points only, one bullet = one mark, no intro/conclusion. Evaluation Mode: estimated score (X/Y), correct points, missing points, mistake type (concept gap / structure issue / missing keywords).

Explanation Mode: analogy → 3-5 plain-English bullets → diagram reference → exam link. Max 150 words, no mark-scoring format.

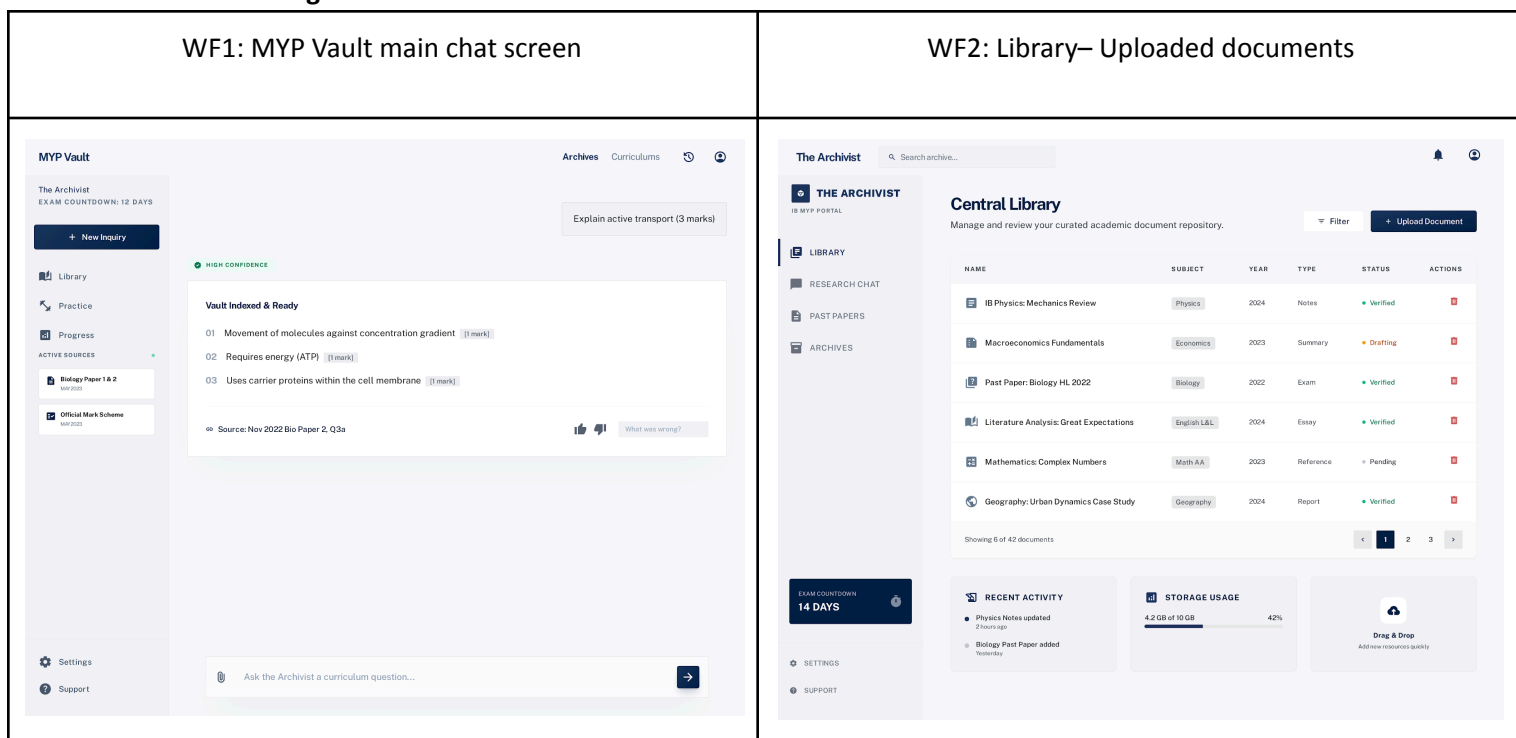
Constraints: must only use retrieved context or credible IB material; must not hallucinate, use outside knowledge, fill gaps with assumptions, or reproduce more than 2 consecutive mark scheme bullets verbatim. Must refuse copyrighted full-document requests and out-of-scope queries.

Full system prompt, functional requirements (MUST/SHOULD/COULD), five worked example interactions, and error-handling table available on request.”

H: Wireframes

Note: Only interface and user flow is displayed. Model screenshots below show output quality for GPT-5 Mini (Exam and Evaluation modes) and Gemini 2.5 Pro (Explanation mode) – each in its intended mode, not a controlled head-to-head. Retrieval logic requires an AI infrastructure that goes beyond UI prototyping tools.

Wireframe using Stitch AI:



Answer Retrieval

Exam Mode (GPT-5 Mini):

"Define osmosis (2 marks)"

- Movement of **water molecules** from a region of **higher water concentration** to **lower water concentration**
- Across a **partially permeable membrane**



Evaluation Mode (GPT-5 Mini):

6c. Consequence 1: there are fewer raccoons for cougars to eat, so the predators' population decreases. Consequence 2: deer population decreases and frogs increase. Grade my answer

• Estimated: 4/5 marks

Correct points:

- Cougar population decreases
- Justification: fewer raccoons as prey
- Deer population decreases
- Frog population increases

Missing points:

- Justification linking frog increase to reduced predation by raccoons
- Justification linking deer decrease to increased predation by cougars

Mistake type:

- Structure issue

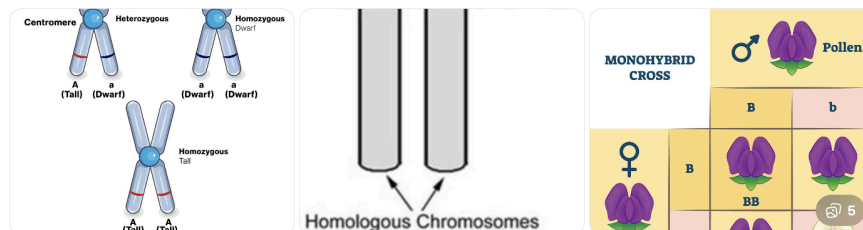
Explanation Mode (Gemini 2.5 Pro):

explain alleles

Analogy: Think of a gene as a recipe and alleles as different versions of that recipe — one from your mum, one from your dad.

Explanation:

- Alleles are different versions of the same gene
- They are found at the same position on homologous chromosomes
- You inherit one allele from each parent
- Dominant alleles are expressed if present
- Recessive alleles are only expressed if both copies are recessive



Exam link: In IB exams, this concept usually appears as genetics questions where you define alleles, identify dominant/recessive traits, and use Punnett squares to predict offspring.